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TE-III

DSBDA



Date

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Unit Test - I

Q. 27

a)

→ The 5 Vs of Big Data are essential characteristics that defines Big data

① Volume

It is the amount of data being generated every second from social media interactions, transaction records, sensor data & more.

② Velocity

speed at which data is generated, processed & analyzed.

The velocity of data affects decision-making.

③ Variety

Types of formats & data including structured data, unstructured data & semi-structured data.

④ Veracity

It is the trustworthiness & quality of the data. Big data can be sometimes be noisy & incomplete.

⑤ Value

The worth of the data in terms of actionable insights. It's not about collecting massive amounts of data, but meaningful data.

b)

→ Data Science	Information Science
i) Analyzing & extracting insights from raw data	i) Managing, organizing, disseminating information
ii) To generate predictions, patterns & insights from data	ii) To ensure efficient storage retrieval & usage of information
iii) Data Mining, ML, Statistics, Big Data	iii) Info Retrieval, dbms, metadata mgmt
iv) Works with large / unstructured / raw db	iv) Works with processed or structured info
ex: Predicting customer behaviour using hist data	ex: Design info systems, organize data

c)

→ Different data types

① Normal Data

This type of data represents categories / names without any order or ranking.
 ex: Gender (male, female)
 Color (R, G, B)

② Ordinal Data

ordinal data represents categories that have a meaningful order but the intervals between them are not necessarily equal
 ex: Educational level

③ Interval Data

it has meaningful differences between values
 ex: Temp in celsius

④ Ratio Data

it also has a true zero point that indicates an absence of quantity
 ex: height, weight, age

Q.37

a2

→ Measures of Central Tendency

a2 Mean

Sum of all values in data divided by number of values

$$\bar{x} = \frac{\sum x_i}{n}$$

ex: [3, 5, 7, 8, 10]

$$\text{mean} = \frac{3+5+7+8+10}{5} = \frac{33}{5} = 6.6$$

b) Median

Middle value when data is ordered from smallest $\rightarrow$ largest

ex: [3, 5, 7, 8, 10]
 ↑
 median

c) Mode

It appears most frequently in data set

ex: [3, 5, 7, 8, 7]
 mode

b)

$\rightarrow$ The Null Hypothesis (H_0) is statement that there is no effect difference or relationship between variables. It is the hypothesis that researchers aim to test against

ex: In study testing whether a new drug has effect on bp

H_0 : The new drug does not affect bp
 here you would test whether data provide sufficient evidence shows different in bp after using drug

It is particularly useful in searching where conditional probability is involved

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

ex: Imagine you are testing for a rare disease
The probability of having the disease
(prior probability $P(A)$ is 0.1% (0.001)

A test for disease has 95% sensitivity & the
5% false positive rate.

Given that a person has ~~test~~ test positive
Bayes theorem helps to calculate the
probability that they actually have disease

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$